

Exercise 3

1. Implement the function X using static CMOS as a one stage “Complex Gate”. A maximum of 12 transistors is allowed.

$$X = \overline{A}\overline{B} + \overline{C}\overline{D} + \overline{E}\overline{D} + \overline{F}\overline{B}$$

2. Implement the function X using static CMOS as a one stage “Complex Gate”. A maximum of 12 transistors is allowed.

$$X = \overline{A}\overline{D} + \overline{B}\overline{D} + \overline{C}\overline{D} + \overline{E}\overline{F}$$

3. Figure shows a simple logic function realized with Boolean gates.

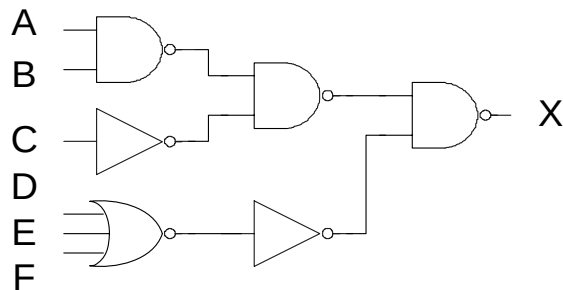


Figure 1: A Boolean function.

Implement the function X using static CMOS as a one stage “Complex Gate”. A maximum of 12 transistors is allowed.

4. Figure 2 shows a simple logic function realized with Boolean gates.

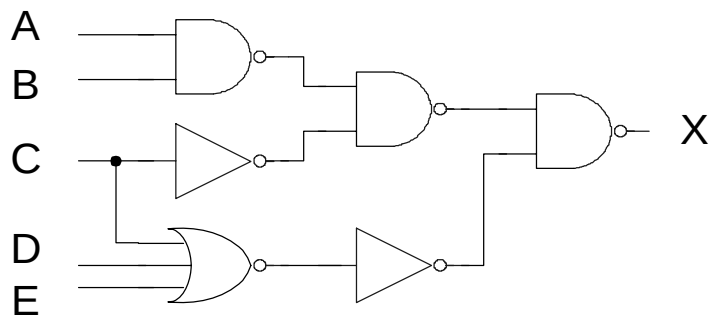


Figure 2: A Boolean function.

Implement the function X using static CMOS as a one stage “Complex Gate”. A maximum of 10 transistors is allowed.

5. Figure 3 shows a simple logic function realized with Boolean gates.

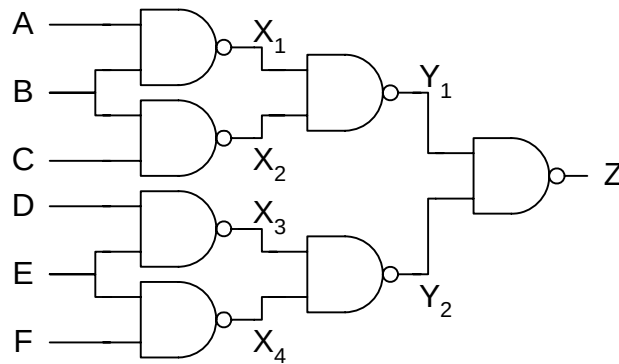


Figure 3: A Boolean function.

Implement the function X using static CMOS as a one stage “Complex Gate”. A maximum of 12 transistors is allowed.

6. Which logic function is shown in Figure 4? Specify both the PDN and PUN function.

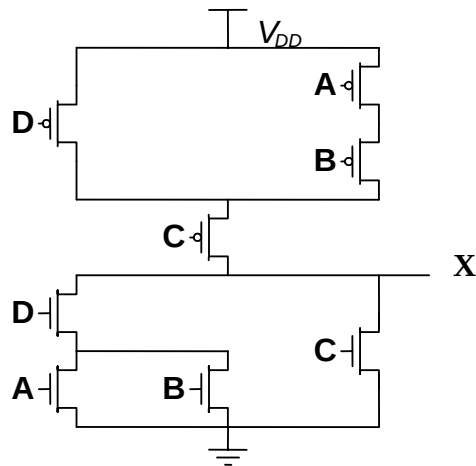


Figure 4: A CMOS circuit with four inputs.

7. To code a word X using Booth's modified algorithm in a bit-sliced array multiplier, shifting and two's complement conversion is needed. The circuit, shown in Figure 5, shows a Booth multiplexer, which often is used for such purpose. The control signal INV is used to choose if the word X should be complemented. The control signals S1 and S2 are used to choose shift X or not and Z is used to set the partial product P to zero.

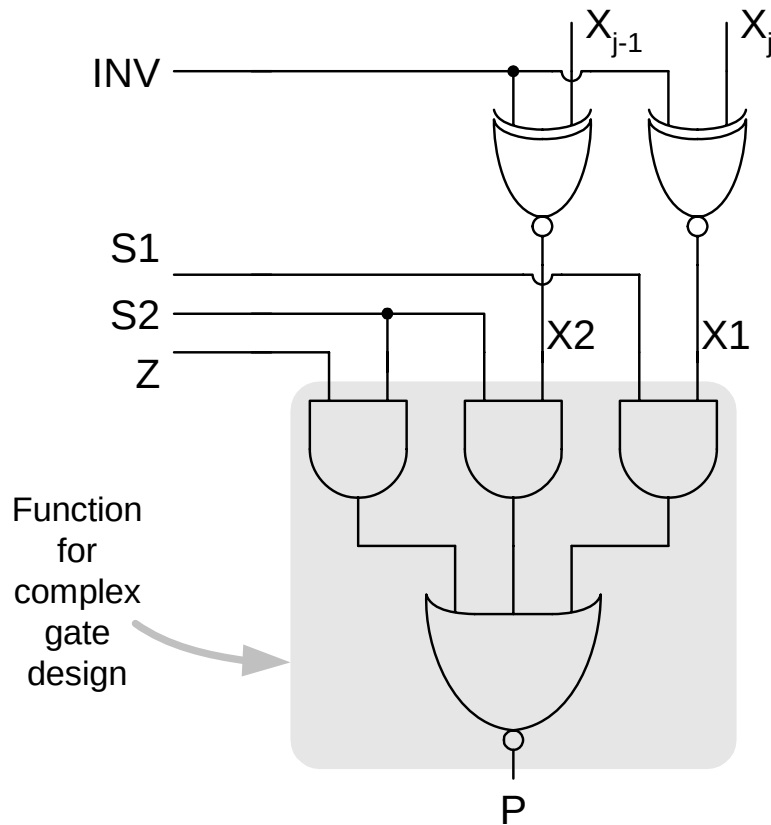


Figure 5: A complex function for Booth’s modified algorithm

The designer decided to try a complex gate design for the part in the gray area in Figure 5. Implement the function $P = f(S1, S2, Z, X1, X2)$ in complementary CMOS as a complex gate. A maximum of 10 transistors can be used.

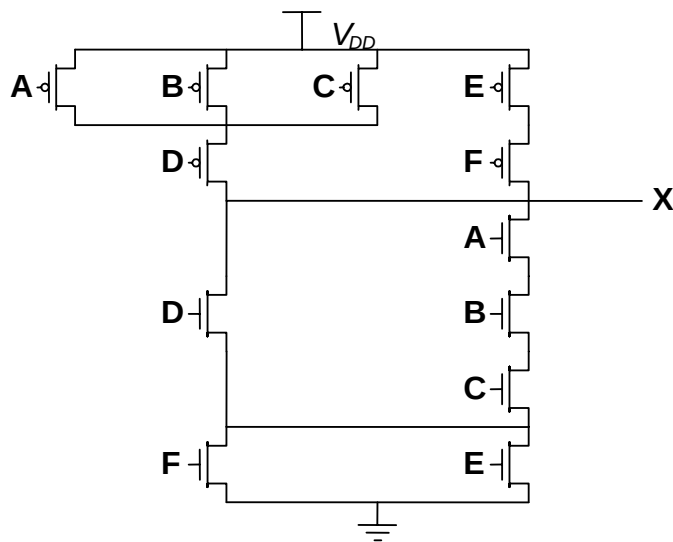


Figure 6: A CMOS circuit with six inputs.

8. Show the two stage logic function of Figure 6, using Boolean symbols as AND, NAND, OR, and NOR.

9. Figure 7 shows a function realized with logic gates.

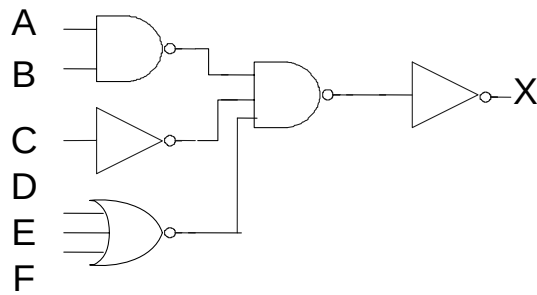


Figure 7: A Boolean function realized with logic gates.

Implement the function X using static CMOS as a one stage “Complex Gate”. A maximum of 12 transistors is allowed.

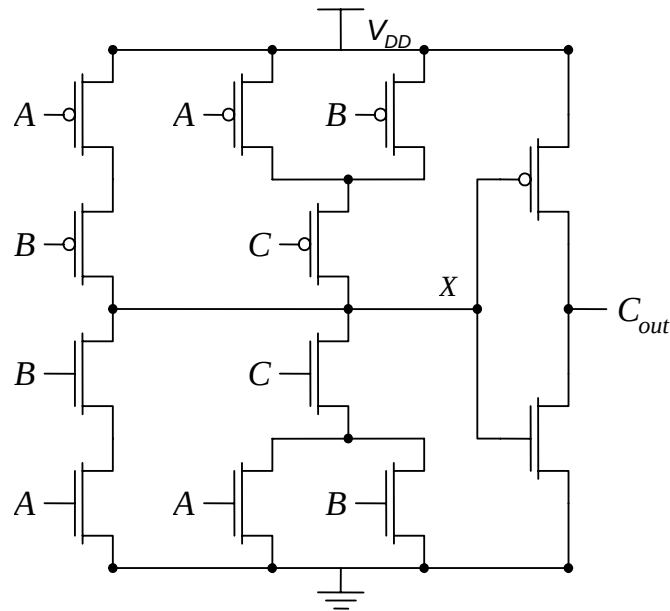


Figure 8: A two stage logic circuit with 3 inputs.

10a. Specify the logic function of the PUN for X in Figure 8. Derive the PDN function from the PUN function, is it the same as shown in Figure 8? What is the function of the last stage in Figure 8?

10b. Modify the schematic such that only 1 transistor is needed in PUN of the first stage.

10c. How is such a logic style called? Give a brief motivation for the main drawback with that logic style.

11a. An unknown CMOS circuit is shown in Figure 9a. Draw the signals B , C , D , E , and Q if A and ϕ are as shown in Figure 9b.

11b. Identify the circuit.

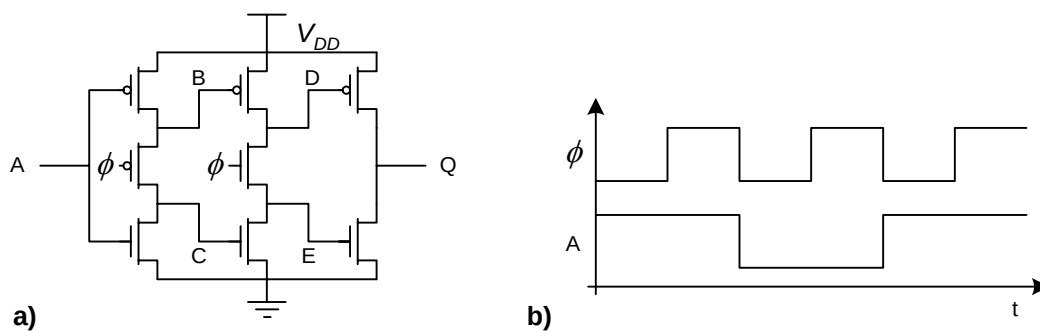


Figure 9: Unknown CMOS circuit.

12. Given is the simplified layout in Figure 7.

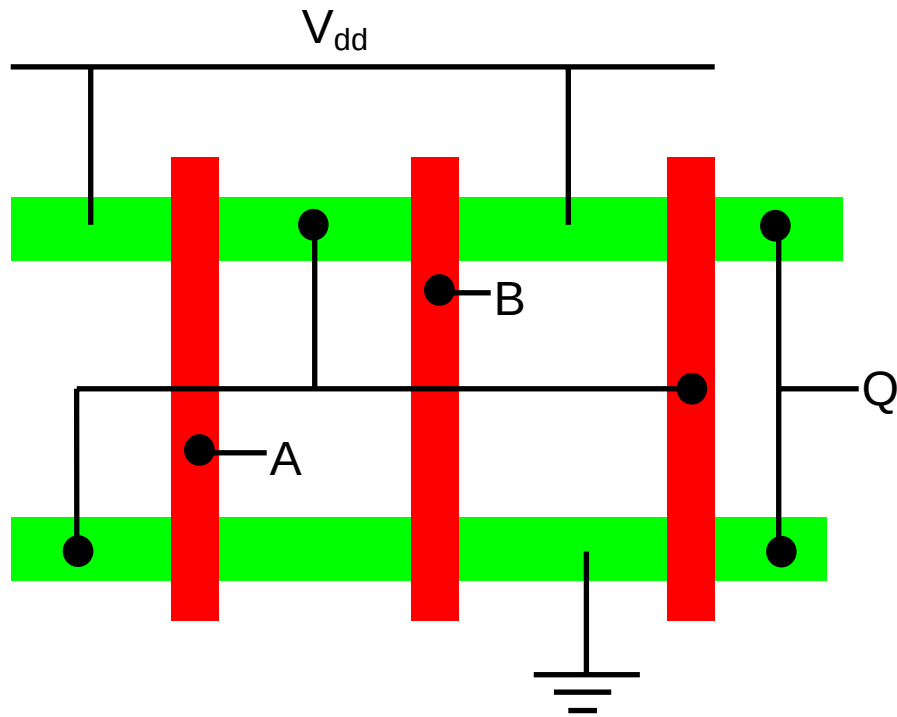


Figure 7: Layout of a Boolean function

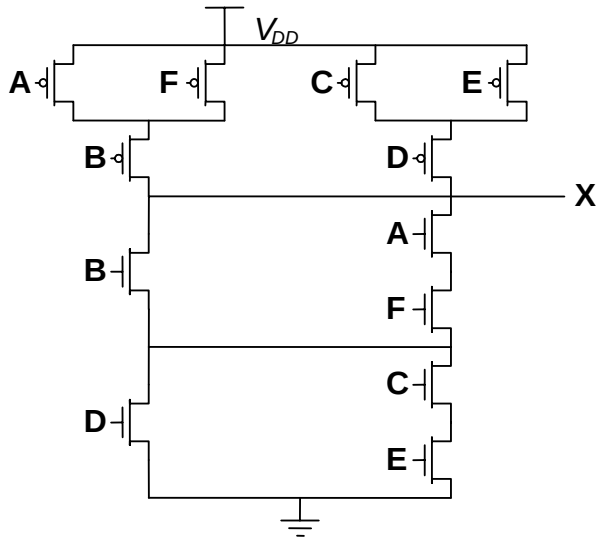
12a. Draw the schematic of the layout. Which operation does the circuit perform?

12b. Mark the Euler paths in the PUN and PDN

Solution 3

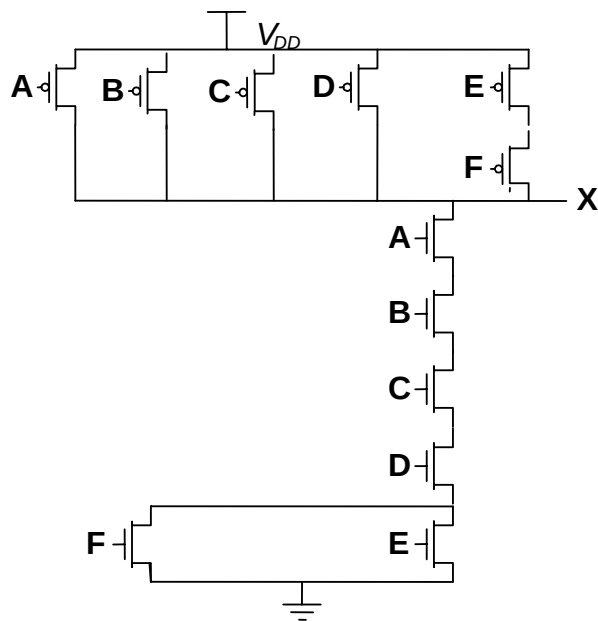
$$1. X = \overline{A} \overline{B} + \overline{C} \overline{D} + \overline{E} \overline{D} + \overline{F} \overline{B} = (\overline{A} + \overline{F}) \overline{B} + (\overline{C} + \overline{E}) \overline{D} ,$$

$$\overline{X} = \overline{(\overline{A} + \overline{F}) \overline{B} + (\overline{C} + \overline{E}) \overline{D}} = (A + F) B + (C + E) D$$

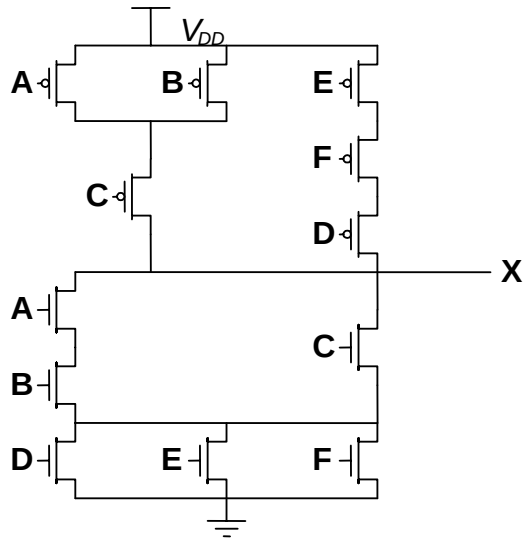


$$2. X = \overline{A} \overline{D} + \overline{B} \overline{D} + \overline{C} \overline{D} + \overline{F} + \overline{E} = \overline{A} + \overline{D} + \overline{B} + \overline{D} + \overline{C} + \overline{D} + \overline{E} \overline{F} = \overline{A} + \overline{D} + \overline{B} + \overline{C} + \overline{E} \overline{F}$$

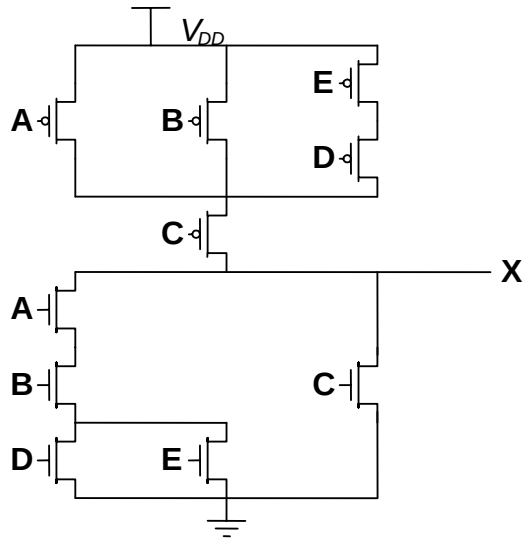
$$\overline{X} = \overline{(\overline{A} + \overline{D})(\overline{B} + \overline{C})(\overline{E} \overline{F})} = A D B C (E + F)$$



$$3. X = \overline{\overline{AB} \overline{C}} (D + E + F) = \overline{(AB + C)(D + E + F)} = (\overline{A} + \overline{B}) \overline{C} + \overline{D} \overline{E} \overline{F} ,$$



$$4. X = \overline{\overline{AB} \overline{C}} (C + D + E) = \overline{(AB + C)(C + D + E)} = (\overline{A} + \overline{B} + \overline{D} \overline{E}) \overline{C} ,$$

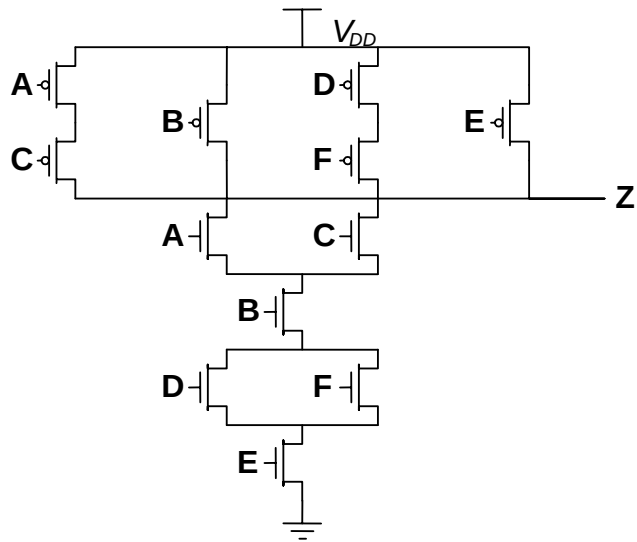


5.

$$Y_1 = \overline{\overline{AB} \cdot \overline{BC}} = AB + BC = B(A + C)$$

$$Y_2 = \overline{\overline{DE} \cdot \overline{EF}} = DE + EF = E(D + F)$$

$$Z = \overline{Y_1 \cdot Y_2} = \overline{B(A + C)E(D + F)}$$



6. $X = (\bar{D} + \bar{A}\bar{B})\bar{C}$ PUN

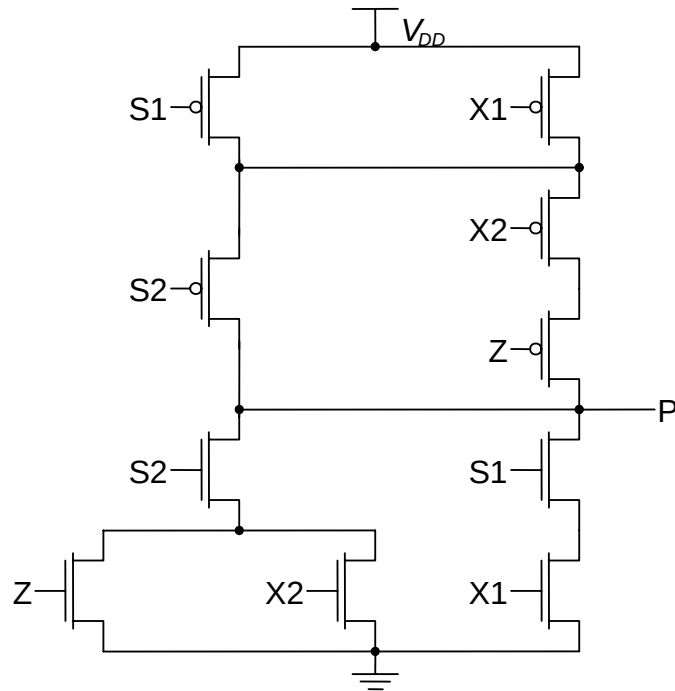
$\bar{X} = \overline{(\bar{D} + \bar{A}\bar{B})\bar{C}} = D(\bar{A}\bar{B}) + C = D(A + B) + C$ PDN

7.

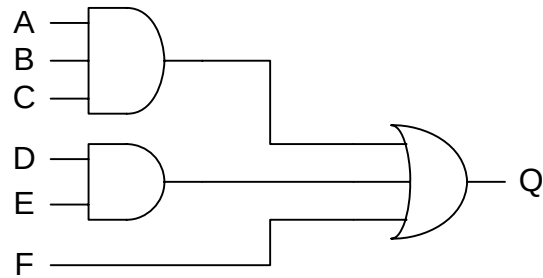
$$P = \overline{Z \cdot S2 + X2 \cdot S2 + X1 \cdot S1} = \overline{(Z + X2) \cdot S2 + X1 \cdot S1}$$

$$f_{PDN} = \bar{P} = (Z + X2) \cdot S2 + X1 \cdot S1$$

$$f_{PUN} = P = (\overline{Z + X2 + S2})(\overline{X1 + S1}) = (\bar{Z} \cdot \bar{X2} + \bar{S2})(\bar{X1} + \bar{S1})$$

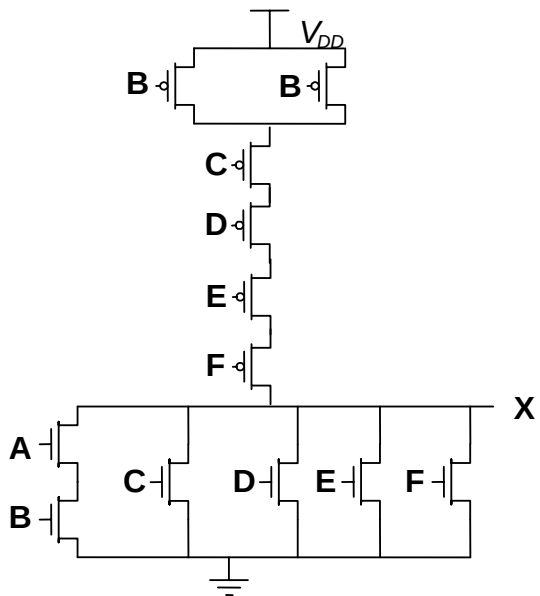


8.



9. The function X is described by $X = \overline{(D + E + F)} \overline{CAB} = \overline{DEFC}(\overline{A + B})$. The equivalent equation for the PDN is determined by De Morgan law's:

$$\overline{X} = \overline{\overline{DEFC}(\overline{A + B})} = D + E + F + C + AB$$

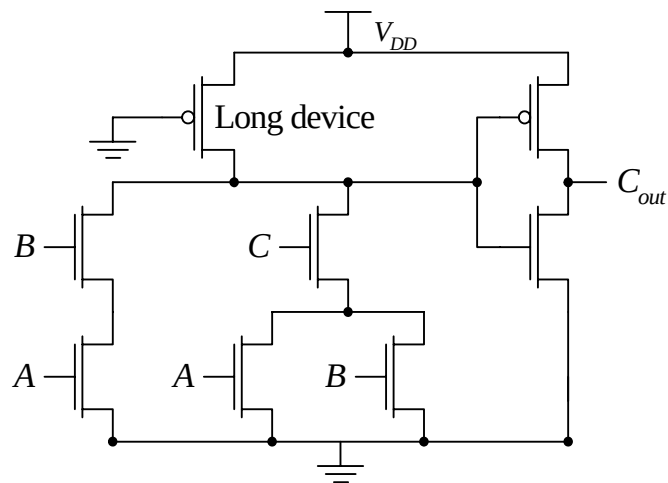


10a. $X = \overline{AB} + (\overline{A} + \overline{B})\overline{C}$
 $\overline{X} = \overline{\overline{AB} + (\overline{A} + \overline{B})\overline{C}} = (A + B)(AB + C) = AAB + ABB + (A + B)C = AB + (A + B)C$

The last stage is an inverter

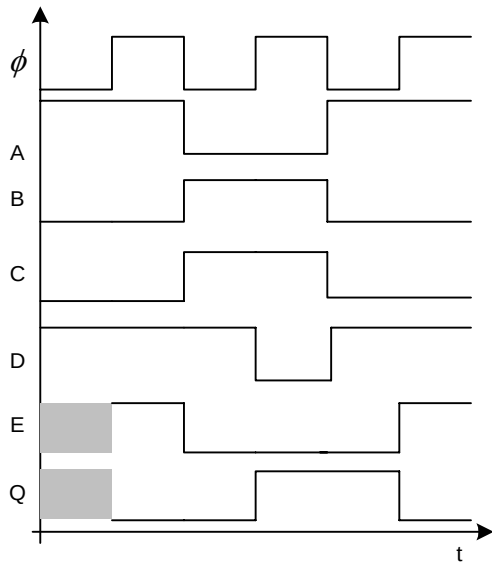
$$C_{out} = \overline{X}$$

10b. Pseudo-NMOS does only need 1 pull up transistor



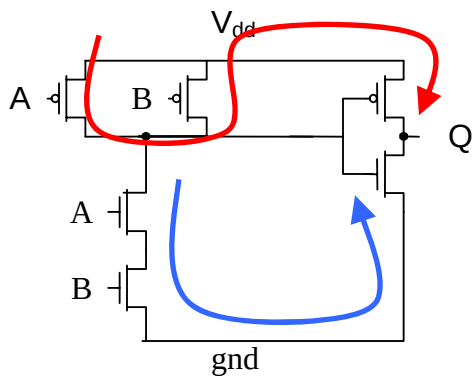
10c. The logic style is called Pseudo-NMOS, which consumes static power.

11a.



11b. A positive edge-triggered D flip-flop using spit-output latches. Q is inverted. It is enough to answer inverting positive edge-triggered flip-flop.

12.



It is an AND-gate.